

MACHINE DESIGN DOCUMENT v1.0

E-Commerce Store Analysis Dashboard

Power BI · DAX · Data Modeling · Business Intelligence

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Author:	Gargeya15
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■ github.com/Gargeya15

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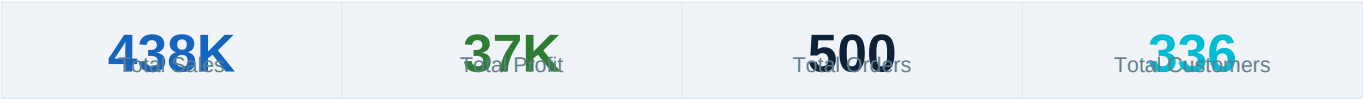
1. Introduction & Purpose

This Machine Design Document (MDD) provides a comprehensive technical and functional specification for the **E-Commerce Store Analysis Dashboard** built using Microsoft Power BI. It serves as the authoritative reference for understanding the system's architecture, data model, DAX logic, visual design, and business requirements.

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Version	1.0 – Initial Release
Date	May 01, 2026
Author	Gargeya15
GitHub	https://github.com/Gargeya15
Scope	End-to-end Power BI dashboard for an e-commerce sales dataset
Audience	Data Analysts, BI Developers, Business Stakeholders

2. Project Overview

The goal of this project is to design and develop a fully interactive Power BI dashboard that allows stakeholders to explore e-commerce performance across four business dimensions: Sales Overview, Product Analysis, Customer Behaviour, and Regional Distribution.



Business Questions Answered:

- Which product categories and sub-categories generate the most revenue and profit?
- Who are the top customers by total sales value?
- Which states and regions contribute most to overall sales?
- How have sales and orders trended over the available time period?
- What payment modes are most preferred by customers?
- Which products have high sales but low or negative profit (loss leaders)?

3. System Architecture & Data Model

The dashboard follows a **Star Schema** data model with one Fact Table and one Dimension Table connected via a one-to-many relationship on **Order ID**.

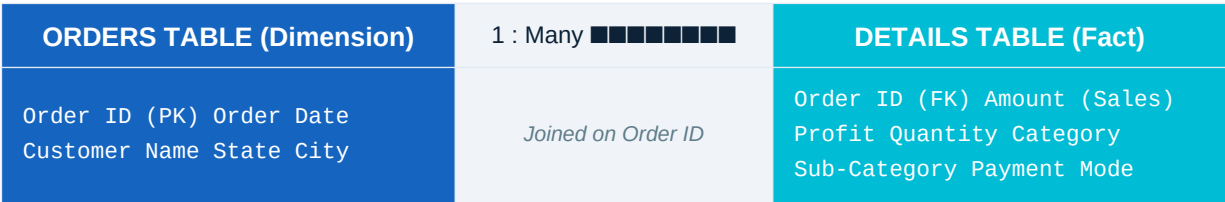


Figure 1 – Star Schema Data Model

Relationship Configuration:

Relationship Type	One-to-Many (1:M)
Join Key	Order ID
Direction	Orders (1) → Details (Many)
Cross Filter	Single direction
Cardinality	One Order ID can appear in multiple Detail rows

4. Dataset Description

Two CSV files form the data source for this project:

4.1 Orders.csv – Dimension Table

Column Name	Data Type	Description	Sample Value
Order ID	Text (PK)	Unique order identifier	B-25601
Order Date	Date	Date order was placed	01-04-2018
Customer Name	Text	Full name of the customer	Harivansh
State	Text	Indian state where order was placed	Uttar Pradesh
City	Text	City of the customer	Mathura

4.2 Details.csv – Fact Table

Column Name	Data Type	Description	Sample Value
Order ID	Text (FK)	Foreign key linking to Orders table	B-25601
Amount	Numeric	Sales amount for the line item (₹)	1800
Profit	Numeric	Profit earned on the line item (₹)	190
Quantity	Integer	Number of units sold	3
Category	Text	Product category	Electronics
Sub-Category	Text	Product sub-category	Printers
Payment Mode	Text	Mode of payment used	COD

4.3 Data Cleaning Steps (Power Query)

- Removed duplicate and null rows from both tables
- Standardised column names (removed spaces, consistent casing)
- Converted Order Date from Text to Date data type
- Converted Amount, Profit, Quantity from Text to Numeric types
- Created derived column: **Month-Year** for trend analysis (e.g., Jan 2018)
- Ensured referential integrity: all Order IDs in Details exist in Orders
- Trimmed whitespace from Category and Sub-Category columns

5. DAX Measures & Calculations

The following DAX measures are defined in the Power BI data model and drive all KPI cards and visual aggregations across the dashboard.

Measure Name	DAX Formula	Business Purpose
Total Sales	SUM(Details[Amount])	Aggregate revenue across all transactions
Total Profit	SUM(Details[Profit])	Aggregate profit across all transactions
Total Quantity	SUM(Details[Quantity])	Total units sold
Total Orders	DISTINCTCOUNT(Details[Order ID])	Count of unique orders
Total Customers	DISTINCTCOUNT(Orders[Customer Name])	Count of unique customers
Avg Order Value	DIVIDE([Total Sales], [Total Orders], 0)	Average revenue per order
Avg Sales per Customer	DIVIDE([Total Sales], [Total Customers], 0)	Average revenue per unique customer

DAX Design Principles Applied:

- Used DIVIDE() instead of '/' to handle zero-division gracefully
- DISTINCTCOUNT() used for Customers and Orders to avoid double-counting
- All measures are stored in a dedicated 'Measures' table for organisation
- Measures are referenced across pages using cross-table DAX context

6. Dashboard Design Specification

6.1 Visual Theme & Style Guide

Primary Color	#1565C0 – Deep Blue (navigation, headers)
Accent Color	#00BCD4 – Cyan (highlights, active states)
Background	#FFFFFF – White (card containers)
Page Background	#F0F4F8 – Off-White
Text Color	#1A2332 – Dark Navy
Font Family	Segoe UI (Power BI default)
Card Style	White containers with soft drop shadow
Navigation Panel	Left-side dark navy panel, icon + label per page
Slicer Style	Dropdown slicers at top-right (State, Category)

6.2 Layout Principles

- Consistent 4-KPI card row at the top of each dashboard page
- Two-column layout for charts (left: trend/scatter, right: ranked bar)

- Map visual full-width when geographic data is the primary focus
- Active navigation icon is highlighted with a circular white background
- Minimum 8px padding inside all card containers
- Responsive visual sizing — charts fill available whitespace evenly

6.3 Accessibility Considerations

- Blue-themed palette maintains WCAG AA contrast ratio on white backgrounds
- All charts include data labels for screen-reader friendliness
- Tooltips enabled on all visuals for additional context on hover
- Alt-text added to map visuals

7. Dashboard Pages – Detailed Design

7.1 ■ Overview Dashboard

High-level executive summary of overall business performance. First page visible on load.

KPI Cards:

- Total Sales: 438K
- Total Profit: 37K
- Total Quantity: 6K
- Total Orders: 500

Visuals:

Visual Title	Chart Type	Description
Sales Trend Over Time	Line Chart	Month-Year on X-axis, Total Sales on Y-axis. Shows declining trend Jan–Jul 2018.
Sales by Payment Mode	Donut Chart	5 segments: COD 35.45%, UPI 19.86%, EMI 17.79%, Credit Card 15.68%, Debit Card 11.22%.
Sales by Category	Bar Chart	Electronics 166K > Clothing 144K > Furniture 127K.
Sales by State	Map Visual	Bubble size proportional to total sales. Uttar Pradesh dominates.

Active Filters / Slicers: State (dropdown slicer), Category (dropdown slicer)

7.2 ■ Products Dashboard

Deep-dive into product-level performance — which products drive revenue and which are loss-makers.

KPI Cards:

- Total Sales: 438K
- Total Profit: 37K
- Total Quantity: 6K
- Avg Order Value: 875.54

Visuals:

Visual Title	Chart Type	Description
Top 10 Selling Products	Horizontal Bar	Ranked by Total Sales. Printers (59K) and Saree (59K) lead; Stole (19K) is 10th.
Sales vs Profit Analysis	Scatter Plot	Each bubble = sub-category. Size = quantity. Identifies high-sales/low-profit outliers.

Active Filters / Slicers: State, Category (cross-filtering from scatter to bar)

7.3 ■ Customers Dashboard

Understand customer value, concentration, and purchasing behaviour.

KPI Cards:

- Total Customers: 336
- Total Sales: 438K
- Total Orders: 500
- Avg Sales/Customer: 1.30K

Visuals:

Visual Title	Chart Type	Description
Top 10 Customers by Sales	Horizontal Bar	Harivansh (9.9K) and Madhav (9.4K) are highest-value customers.
Orders by Category	Donut Chart	Clothing 50.19% (393 orders), Electronics 26.05% (204), Furniture 23.75% (186).
Customer Orders Trend	Line Chart	Monthly order count Jan–Sep 2018. Peak in Jan (60+), declining toward Sep.

Active Filters / Slicers: State, Category

7.4 ■ Regions Dashboard

Analyse geographic distribution of sales and identify top-performing states.

Visuals:

Visual Title	Chart Type	Description
Sales Distribution by State	Map Visual	Large bubble map centred on India. Bubble = Total Sales by state.
Top Sales by State	Horizontal Bar	UP 31K > Maharashtra 17K > Madhya Pradesh 13K > Delhi 6K > Karnataka 1K.
Customer Orders Trend	Line Chart (cat)	Total Sales on Y-axis against Category on X-axis — Electronics highest.

Active Filters / Slicers: State, Category

8. Interactivity & Filter Behaviour

Feature	Type	Behaviour
State Slicer	Dropdown	Filters all visuals on current page to selected state(s)
Category Slicer	Dropdown	Filters visuals to selected product category
Cross-Filter (Charts)	Bi-directional	Clicking a bar/segment filters other visuals on the same page
Drill-Down	Hierarchy	Date hierarchy (Year > Quarter > Month) on trend charts
Tooltip	Hover	All visuals show detailed values on mouse hover
Navigation Panel	Button	Left-side icons navigate between the 4 dashboard pages
Reset Filters	Clear All	Top-level slicer clear button resets all selections

9. Data Cleaning & Transformation

St ep	Operation	Tool	Outcome
1	Remove duplicates	Power Query	Ensures each Order ID is unique in Orders table
2	Handle null values	Power Query	Null rows dropped; no imputation applied
3	Data type conversion	Power Query	Date, Numeric, Text types enforced on import
4	Column rename	Power Query	Standardised to PascalCase naming convention
5	Derived column: Month-Year	Power Query	Format: MMM YYYY — used as X-axis on trend charts
6	Referential integrity	Power Query	Merge query validates all FK Order IDs exist in Orders
7	Whitespace trim	Power Query	Text.Trim() applied to Category and Sub-Category

10. Key Business Insights

Insight	Detail
Electronics dominates	Electronics contributes 166K in sales (37.9% of total), making it the highest revenue category.
Loss-leader products	Scatter plot reveals certain sub-categories have high sales volume but negative or near-zero profit.
Customer concentration	Top 2 customers (Harivansh 9.9K, Madhav 9.4K) account for ~4.4% of total revenue combined.
Regional skew	Uttar Pradesh alone contributes 31K — nearly 7.1% of all sales, far exceeding other states.
COD preference	35.45% of orders use Cash on Delivery, indicating trust issues with digital payments or rural customer base.
Declining order trend	Monthly orders declined from ~60 in Jan 2018 to ~30 by Sep 2018, warranting further investigation.

Insight	Detail
Clothing volume leader	Clothing has the highest order count (393 = 50.19%) despite lower per-unit value than Electronics.

11. Non-Functional Requirements

Requirement	Category	Target / Specification
Report Load Time	Performance	< 5 seconds on standard hardware with full dataset
Dataset Size	Scalability	Tested with 500 orders; scalable to 50,000+ rows
Browser Compatibility	Compatibility	Power BI Service: Chrome, Edge, Firefox
Mobile Responsiveness	Usability	Power BI mobile layout configured for portrait view
Access Control	Security	Row-level security (RLS) can be applied by State
Refresh Schedule	Freshness	Manual refresh; automated via Power BI Gateway if needed
Version Control	Maintenance	Source CSV files version-controlled on GitHub

12. Risk & Mitigation

Risk	Severity	Mitigation Strategy
Data type mismatch on import	High	Power Query enforces explicit data types on each column
Missing Order IDs in Details	High	Referential integrity check performed in Power Query merge
Negative profit values misread	Medium	Data validated; negative profit = intentional loss-leader data
Slicer cross-filter conflicts	Medium	Visual interaction settings reviewed and configured per page
Performance lag with large dataset	Low	Aggregations pre-computed via DAX measures; no row-level iteration
Incorrect Month-Year sort order	Low	Sort by numeric month column, not alphabetical Month-Year string

13. Glossary

Term	Definition
MDD	Machine Design Document — formal technical specification for a BI/analytics system
DAX	Data Analysis Expressions — formula language used in Power BI, Excel, and SSAS
Power Query	ETL tool built into Power BI for data ingestion, cleaning, and transformation
Star Schema	Data modelling pattern with central Fact table surrounded by Dimension tables
KPI	Key Performance Indicator — measurable metric used to evaluate business performance
Fact Table	Central table containing transactional, measurable data (Details.csv)
Dimension Table	Descriptive table providing context for facts (Orders.csv)
COD	Cash on Delivery — payment collected at time of product delivery
RLS	Row-Level Security — restricts data visibility per user role in Power BI
Slicer	Interactive filter visual in Power BI allowing users to segment data
Cross-filter	Interaction where selecting data in one visual filters data in another

14. References & Links

GitHub Repository	https://github.com/Gargeya15
Dataset – Orders.csv	https://github.com/Gargeya15 (Orders.csv)
Dataset – Details.csv	https://github.com/Gargeya15 (Details.csv)
Power BI Desktop	https://powerbi.microsoft.com/desktop

DAX Reference	https://docs.microsoft.com/en-us/dax/
Power Query Docs	https://docs.microsoft.com/en-us/power-query/
ReportLab (PDF lib)	https://www.reportlab.com/docs/reportlab-userguide.pdf

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